SIMPLE TEXT EDITOR LOGIC PRESENTATION

JAVA PROJECT

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JAVA PROJECT BASED ON OOPS PRINCIPLES

- THE 4 OOPS PRINCIPLES ARE:
- DATA ABSTRACTION
- ENCAPSULATION
- INHERITANCE
- POLYMORPHISM

INTRODUCTION

A text editor is a software application used to create, edit, and manage textual content. Text editors are widely used in programming, documentation, note taking, and content creation.

This project focuses on developing the core logic of a simple text editor using Java and Object-Oriented Programming (OOP) principles.

The project demonstrates:

- Classes and Objects
- Encapsulation
- StringBuffer (Mutable Strings)
- Method Design
- File Handling

OBJECTIVE OF THE PROJECT

- To understand Object-Oriented Programming concepts.
- To implement encapsulation and data hiding.
- To learn the use of StringBuffer for mutable string operations.
- To design methods for text manipulation.
- To build a simple real-world application using Java.

WORKING OF THE PROJECT

- The project consists of a Document Class that stores text using a private StringBuffer object.

Step 1: Create Class

- Represents the text editor.
- Manages all text operations.

Step 2: Initialize StringBuffer

- Stores text internally.
- Prevents direct access from outside the class.

Step 3: Constructor

- Initializes an empty text buffer.

Step 4: Add Text

- Appends user-entered text.

Step 5: Clear Text

Removes all existing text

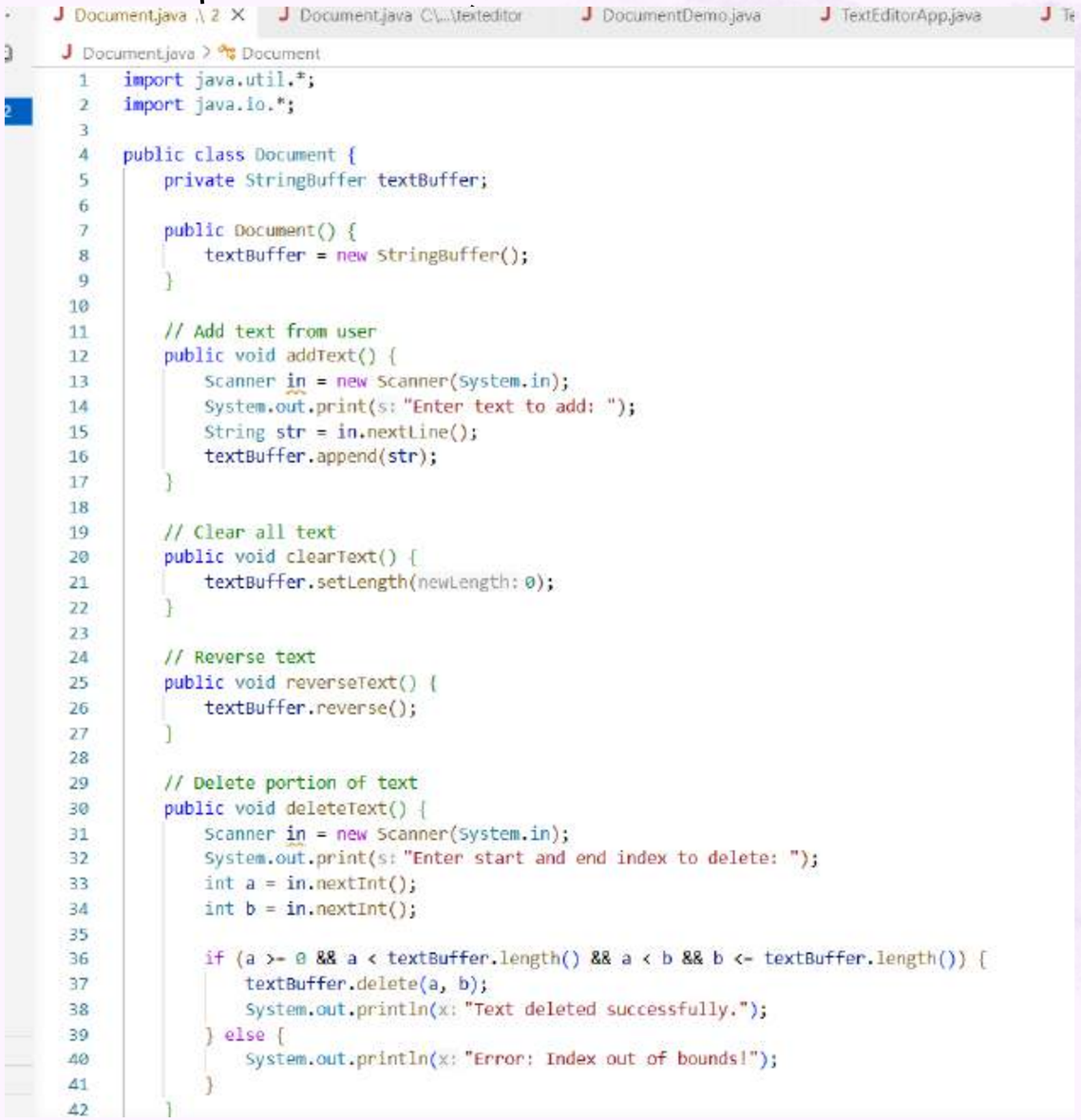
Step 6: Reverse Text

- Reverses the current content.

Step 7: Display Text

- Shows the current text stored in the editor.

SCREENSHOTS



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```
80      public static void main(String[] args) {
93          doc.displayText();
94
95          // Save to file
96          System.out.print(s: "\nEnter filename to save (e.g., mytext.txt): ");
97          String saveFile = sc.nextLine();
98          doc.saveToFile(saveFile);
99
100         // Clear and load from file
101         System.out.println(x: "\nClearing current text...");
102         doc.clearText();
103         doc.displayText();
104
105         System.out.println(x: "Loading from file...");
106         doc.loadFromFile(saveFile);
107         doc.displayText();
108
109         // Optional: Delete operation
110         System.out.print(s: "\nDo you want to delete some text? (y/n): ");
111         String choice = sc.nextLine();
112         if (choice.equalsIgnoreCase(anotherString: "y")) {
113             doc.deleteText();
114             doc.displayText();
115         }
116         System.out.print(s: "\nEnter filename to save (e.g., mytext.txt): ");
117         saveFile = sc.nextLine();
118         doc.saveToFile(saveFile);
119
120         System.out.println(x: "Loading from file...");
121         doc.loadFromFile(saveFile);
122         doc.displayText();
123         sc.close();
124         System.out.println(x: "\n=== Program Ended ===");
125     }
126 }
```

OUTPUT

```
Enter filename to save (e.g., mytext.txt): MyText.txt
Text saved successfully to: MyText.txt

Clearing current text...
Current Content:
Loading from file...
 Text loaded successfully from: MyText.txt
Current Content: lrig a ma I


Do you want to delete some text? (y/n): Y
Enter start and end index to delete: 3
6
Text deleted successfully.
Current Content: lri ma I


Enter filename to save (e.g., mytext.txt): text.txt
Text saved successfully to: text.txt
Loading from file...
 Text loaded successfully from: text.txt
Current Content: lri ma I


=== Program Ended ===
```

ADVANTAGES OF THE PROJECT

- Simple and easy to understand.
- Demonstrates practical application of OOP.
- Efficient text manipulation using StringBuffer.
- Encourages modular programming.
- Provides a foundation for advanced text editor development.

FUTURE SCOPE

- The project can be enhanced by adding:
- Save and Open File functionality
- Undo and Redo operations
- Find and Replace feature
- Word and Character Count
- Graphical User Interface (GUI) using Java Swing
- Font and Text Formatting Options
- Spell Checking System
- Multi-document support
- These improvements can transform the project into a more advanced text editing application.

CONCLUSION

The Simple Text Editor project successfully demonstrates the implementation of Object-Oriented Programming concepts in Java. By using encapsulation, classes, objects, and StringBuffer operations, the project provides an efficient way to manage and manipulate text. It improves understanding of Java programming and illustrates how real-world software applications are developed using OOP principles.

THANK YOU

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